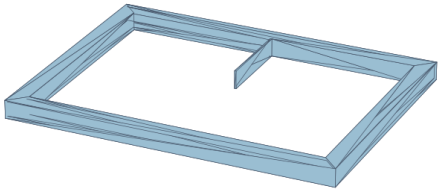
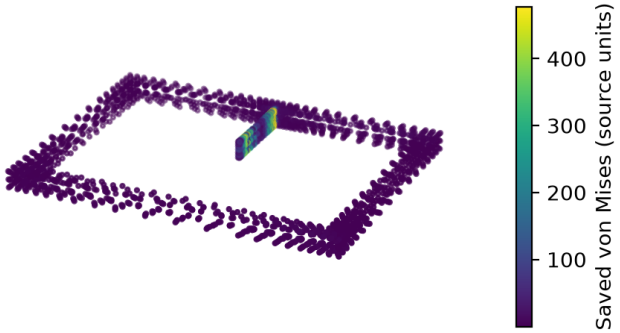
Published test-bench frame CAD + FEM

12 October 2026 | Web-source collection extension | No local CAD creation, FEM solves or model calls

Added one native CAD case with saved numerical fields and one kiteboard CAD candidate with published FEM plots/scalar. Neither is benchmark-certified. Original dimensioned drawing coverage: zero for these two sources.



Original CAD inspection: valid FreeCAD BREP, one solid and 26 faces; bounding dimensions 2000 x 1500 x 100 in native coordinates. The model retains its sketch, pad, pocket and Boolean history.



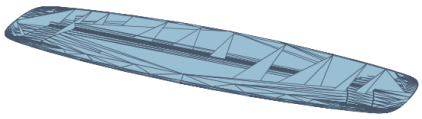
Saved CalculiX reference: stress values at original mesh coordinates. All 2356 nodes have matched source IDs, finite displacement/stress/strain values, and consistent displacement magnitudes. Maximum |u| = 19.0097107742; maximum von Mises = 477.6683112864, in source units.

Von Mises recomputed only as a consistency check on the saved stress tensor: maximum absolute difference 1.14e-13. No FEM solution generated. Steel E = 200000 MPa, Poisson ratio = 0.3. Native setup includes self-weight, a fixed face, and prescribed X displacement -19. Unit/replay and mesh-convergence checks remain before admission.

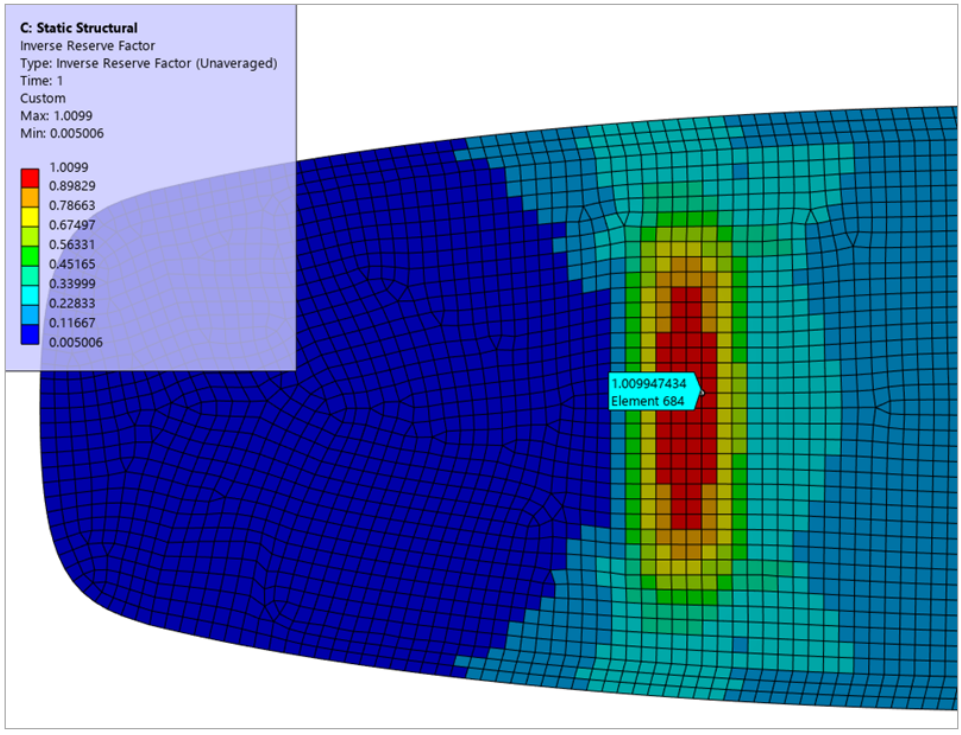
Primary source: MarcusAdielsonGuichet / Test-Bench-Digital-Twin. No explicit data license located. The four external small-mesh FRD test runs are unpaired supporting files, not additional frame cases.

Composite kiteboard: CAD + published FEM reference

Original Ansys ACP/Mechanical workshop download contains STEP shell geometry, a separate solid core and native Workbench project archives. STEP core: one valid solid, 46 faces. Shell STEP: three faces, zero solids. This adds a shaped composite product and failure-analysis task.



CAD inspection render: original kiteboard_core.stp from Kiteboard_SOLVED_2025R1. No geometry generated. The shell and core remain separate original files.



Original published FEM figure: maximum inverse reserve factor 1.0099; shown probe reads 1.009947434 at element 684. This is a dimensionless failure-factor reference, not a frontier-model score.

The download explicitly skips file.rst, ds.dat and solve.out. Keep this as a plot/scalar-reference candidate; no raw nodal fields are present. The reference page is v261 and its linked archive 2025R1: version/setup correspondence and scoring tolerance require review. Replay needs Ansys ACP/Mechanical; no COMSOL dependency.

Sources: original archive setup page; published FEM postprocessing references. Publisher copyright retained; no standalone data license asserted.

Collection inventory and benchmark use

Two new task families: structural frame deformation/stress and composite kiteboard failure. Candidate labels preserve differences in reference completeness. The source files and extracted field table are supplied for local research review.

Source	CAD	FEM reference	Status
Test-bench frame	Native FreeCAD BREP	2356 nodal vectors + stress/strain	Candidate, raw fields verified
Composite kiteboard	Original STEP shell + solid core	Published plots; IRF scalar 1.0099	Candidate, no raw field file
Four other ACP workshops	Native project geometry / auxiliary STEP	No raw fields verified	Held; excluded from count

Counts: one native CAD + saved numerical-field case; one native CAD + published plot/scalar candidate; zero newly certified benchmark cases. Saved solver outputs, mesh nodes, plies and CAD auxiliary tools are not counted as independent tasks.

Track readiness: supplied-CAD-to-FEM candidates await reproducibility, source consistency and tolerance review. Drawing-to-CAD-to-FEM receives no complete dimensioned-drawing case in this extension. CAD inspection renders do not replace original engineering drawings.

Held sources: bonded T-joint, advanced sandwich panel, composite solid extrusion and cutoff-rule modeling. Their original archives were downloaded and inventoried under web-cad-fem-acp; no raw .rst/.rth fields verified. A modeling tutorial marked SOLVED is not evidence of a saved FEM solution. A fusion graphite-foam heat-exchanger lead has numerical temperature profiles and a hybrid FE mesh, but no native CAD/BREP found in the linked record.

Package: original native CAD/archives, untouched source result files, source links and revisions, source image references, SHA256 manifest, node-ID-preserving extracted frame CSV, and audit catalog. No downloaded code or solver was executed. No frontier-model performance measured.

Source handling: source copyright/data-license status recorded in catalog. Local collection only; no repository commit or public release performed. Original downloads preserved separately from derived inspection figures and extracted values.